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5. The chart describes the laboratory preparation of ethanol from glucose solution.

Process **A**		Process **B**		Process **C**
Yeast is added to glucose solution and left in a warm place until the reaction stops	→	Unreacted yeast is removed from ethanol solution	→	Ethanol is separated from the solution

The equation for the reaction occurring in process **A** is shown below.

$$C_6H_{12}O_6(aq) \longrightarrow 2C_2H_5OH(aq) + 2CO_2(g)$$

(a) (i) Give the name of the processes **A**, **B** and **C**. [3]

A ....

B ....

C ....

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(ii) A teacher wanted to show that ethanol is collected in process **C**.

I. Tick (✓) the box next to the chemical test that the teacher would carry out to positively identify the liquid as ethanol. [1]

add bromine water ☐

add acidified potassium dichromate solution ☐

add silver nitrate solution ☐

add barium chloride solution ☐

II. Tick (✓) the box next to the observation you would expect. [1]

orange to colourless ☐

orange to green ☐

green to orange ☐

colourless to green ☐

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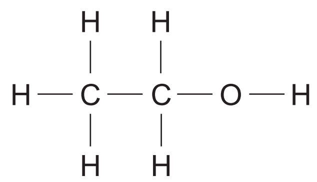
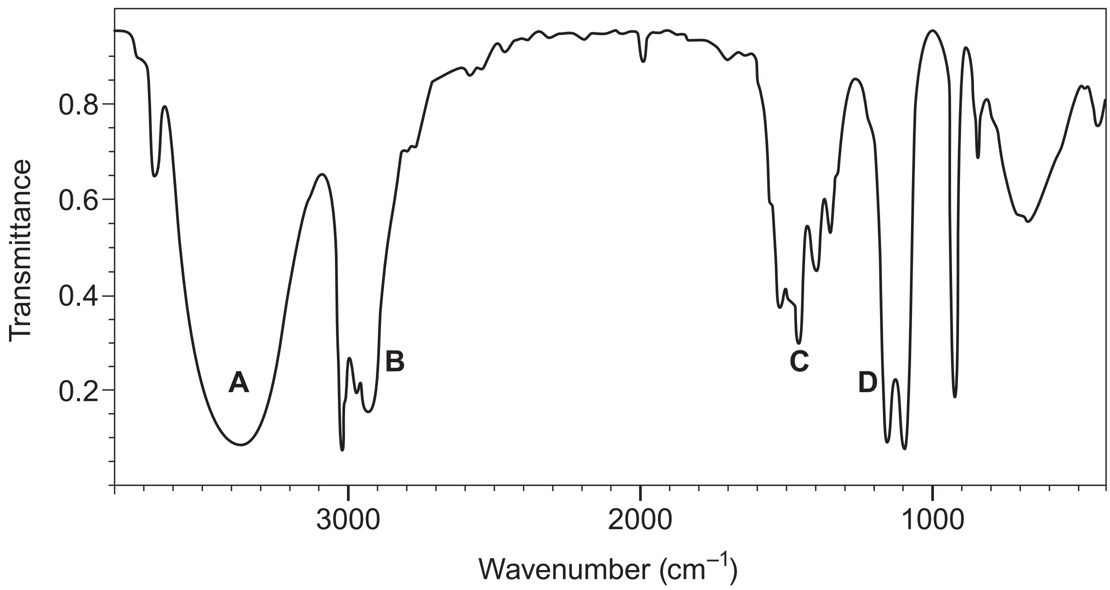
(b) Alcohols can also be identified using infrared spectroscopy.

Bond	Wavenumber (cm^{-1})
C═C	1620 to 1670
C═O	1650 to 1750
C—H	2800 to 3100
O—H	2500 to 3550

(i) The structural formula and the infrared spectrum of ethanol are shown.

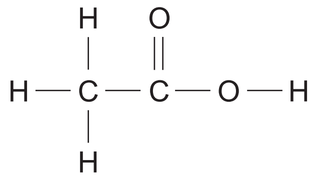


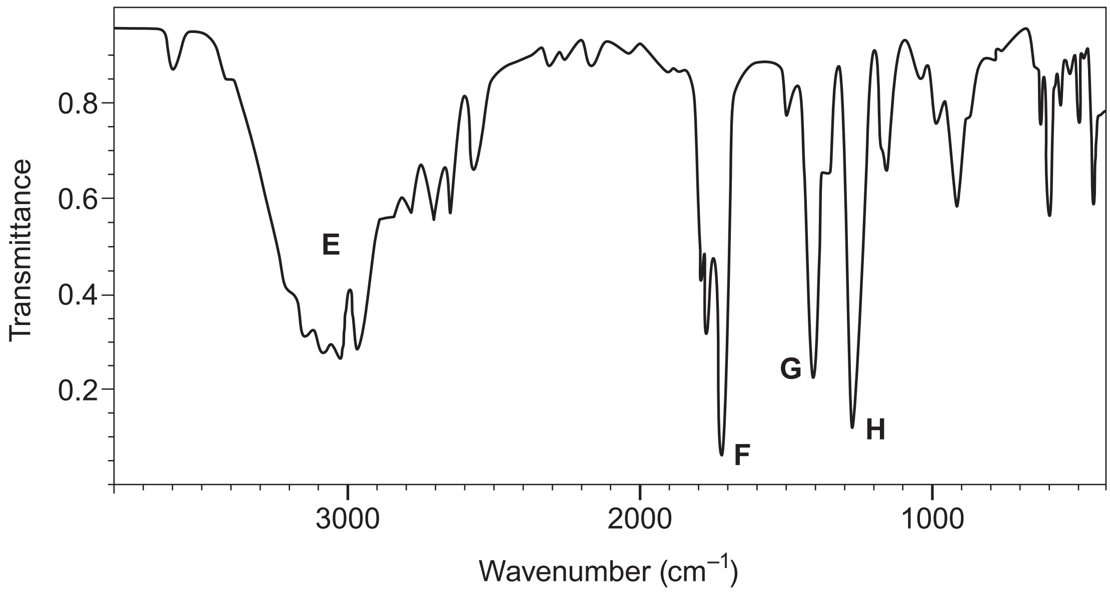


Give the **letter** of the peak that can be used to identify an alcohol. [1]

Letter ....

(ii) The structural formula and the infrared spectrum of ethanoic acid are shown.





Give the **letter** of the peak that can be used to distinguish ethanoic acid from ethanol. [1]

Letter ....

(c) Ethanol is found in alcoholic drinks.

(i) Give **one** health problem associated with alcohol abuse over a **long** period of time. [1]

....

(ii) Give **one** social problem associated with the excessive intake of alcohol (binge drinking) during an evening. [1]

....

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